Computer Organization and Architecture (CSL251)
LAB-1
2025-26 W

Installing the Spike RISC-V Simulator for Linux

Update the Package List:

sudo apt update

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~$ sudo apt update
[sudo] password for deepesh:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,391 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,684 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [363 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [566 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [311 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main Icons (48x48) [36.0 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main Icons (64x64) [51.0 kB]
```

Install Build and Development Dependencies:

sudo apt-get install autoconf automake autotools-dev curl python3 libmpc-dev libmpfr-dev libgmp-dev gawk build-essential bison flex texinfo gperf libtool patchutils bc zlib1g-dev libexpat-dev

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~$ sudo apt-get install autoconf automake autotools-dev curl python3 libmpc-dev libmpfr-dev libgmp-dev gawk build-essential bison flex texinfo gperf
 libtool patchutils bc zlib1g-dev libexpat-dev
[sudo] password for deepesh:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'libexpat1-dev' instead of 'libexpat-dev'
autoconf is already the newest version (2.71-3).
automake is already the newest version (1:1.16.5-1.3ubuntu1).
autotools-dev is already the newest version (20220109.1).
curl is already the newest version (8.5.0-2ubuntu10.6).
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3 set to manually installed.
libmpc-dev is already the newest version (1.3.1-1build1.1).
libmpfr-dev is already the newest version (4.2.1-1build1.1).
libgmp-dev is already the newest version (2:6.3.0+dfsg-2ubuntu6.1).
gawk is already the newest version (1:5.2.1-2build3).
build-essential is already the newest version (12.10ubuntu1).
```

Set Up Device Tree Support Package:

sudo apt install device-tree-compiler

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~$ sudo apt install device-tree-compiler
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libfdt1
The following NEW packages will be installed:
  device-tree-compiler libfdt1
0 upgraded, 2 newly installed, 0 to remove and 265 not upgraded.
Need to get 248 kB of archives.
After this operation, 485 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libfdt1 amd64 1.7.0-2build1 [20.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 device-tree-compiler amd64 1.7.0-2build1 [228 kB]
Fetched 248 kB in 3s (92.5 kB/s)
Selecting previously unselected package libfdt1:amd64.
```

Create Working Directory for RISC-V Setup:

mkdir RISCV
cd RISCV

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~$ mkdir RISCV
deepesh@deepesh-HP-Laptop-15-da0xxx:~$ cd RISCV
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$
```

Clone the repositories:

git clone --recursive https://github.com/riscv/riscv-gnu-toolchain

(This is 6 GB long, so it may take some time. Depending on internet speed and your system configuration. You can also download the repository from the github and extract manually.)

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$ git clone --recursive https://github.com/riscv/riscv-gnu-toolchain
Cloning into 'riscv-gnu-toolchain'...
remote: Enumerating objects: 10117, done.
remote: Counting objects: 100% (32/32), done.
remote: Compressing objects: 100% (23/23), done.
remote: Total 10117 (delta 16), reused 9 (delta 9), pack-reused 10085 (from 3)
Receiving objects: 100% (10117/10117), 6.78 MiB | 160.00 KiB/s, done.
Resolving deltas: 100% (5147/5147), done.
Submodule 'binutils' (https://github.com/bminor/binutils-gdb.git) registered for path 'binutils'
Submodule 'dejagnu' (https://git.savannah.gnu.org/git/dejagnu.git) registered for path 'dejagnu'
Submodule 'gcc' (https://github.com/gcc-mirror/gcc.git) registered for path 'gcc'
Submodule 'gdb' (https://github.com/bminor/binutils-gdb.git) registered for path 'gdb'
Submodule 'glibc' (https://github.com/bminor/glibc.git) registered for path 'glibc'
Submodule 'llvm' (https://github.com/llvm/llvm-project.git) registered for path 'llvm'
Submodule 'musl' (https://git.musl-libc.org/git/musl) registered for path 'musl'
Submodule 'newlib' (https://github.com/bminor/newlib.git) registered for path 'newlib'
Submodule 'pk' (https://github.com/riscv-software-src/riscv-pk.git) registered for path 'pk'
Submodule 'qemu' (https://gitlab.com/qemu-project/qemu.git) registered for path 'qemu'
```

git clone https://github.com/riscv/riscv-pk

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$ git clone https://github.com/riscv/riscv-pk
Cloning into 'riscv-pk'...
remote: Enumerating objects: 4370, done.
remote: Counting objects: 100% (249/249), done.
remote: Compressing objects: 100% (55/55), done.
remote: Total 4370 (delta 220), reused 195 (delta 194), pack-reused 4121 (from 3)
Receiving objects: 100% (4370/4370), 1.60 MiB | 1.05 MiB/s, done.
Resolving deltas: 100% (3113/3113), done.
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$
```

git clone https://github.com/riscv/riscv-isa-sim

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$ git clone https://github.com/riscv/riscv-isa-sim
Cloning into 'riscv-isa-sim'...
remote: Enumerating objects: 26965, done.
remote: Counting objects: 100% (1434/1434), done.
remote: Compressing objects: 100% (450/450), done.
remote: Total 26965 (delta 1133), reused 987 (delta 984), pack-reused 25531 (from 3)
Receiving objects: 100% (26965/26965), 7.33 MiB | 268.00 KiB/s, done.
Resolving deltas: 100% (17898/17898), done.
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$
```

Configure RISCV Environment Variables:

nano ~/.bashrc

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$ nano ~/.bashrc
```

Add the following lines at the end of the file:

export RISCV=/home/deepesh/RISCV
export PATH=$PATH:$RISCV/bin

```
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi
export RISCV=/home/deepesh/RISCV
export PATH=$PATH:$RISCV/bin
```

Press Ctrl + O → press Enter to save
Press Ctrl + X → to exit the editor

Reload the configuration:

source ~/.bashrc

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$ source ~/.bashrc
```

Build the toolchain:

cd riscv-gnu-toolchain

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$ cd riscv-gnu-toolchain
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain$
```

mkdir build
cd build

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain$ mkdir build
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain$ cd build
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain/build$
```

../configure --prefix=$RISCV

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain/build$ ../configure --prefix=$RISCV
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes
checking for gcc option to enable C11 features... none needed
checking for grep that handles long lines and -e... /usr/bin/grep
checking for fgrep... /usr/bin/grep -F
checking for grep that handles long lines and -e... (cached) /usr/bin/grep
checking for bash... /bin/bash
checking for __gmpz_init in -lgmp... yes
checking for mpfr_init in -lmpfr... yes
```

Build the Project Using Make:

make

(This will take some time)

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain/build$ make
mkdir -p /home/deepesh/RISCV/.test || \
        (echo "Sorry, you don't have permission to write to" \
         "'/home/deepesh/RISCV'. Please make sure that the location is " \
         "writable or use --prefix to specify another destination.'" \
         && exit 1)
rm -r /home/deepesh/RISCV/.test
mkdir -p stamps/ && touch stamps/check-write-permission
rm -rf stamps/build-binutils-newlib build-binutils-newlib
mkdir build-binutils-newlib
cd build-binutils-newlib && CC_FOR_TARGET=riscv64-unknown-elf-gcc /home/deepesh/RISCV/riscv-gnu-toolc
        --target=riscv64-unknown-elf \
         \
        --prefix=/home/deepesh/RISCV \
        --enable-plugins \
         \
```

Verify RISC-V Cross Compiler Path and Version:

riscv64-unknown-elf-gcc -v
which riscv64-unknown-elf-gcc -v

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain/build$ riscv64-unknown-elf-gcc -v
Using built-in specs.
COLLECT_GCC=riscv64-unknown-elf-gcc
COLLECT_LTO_WRAPPER=/home/deepesh/RISCV/libexec/gcc/riscv64-unknown-elf/15.2.0/lto-wrapper
Target: riscv64-unknown-elf
Configured with: /home/deepesh/RISCV/riscv-gnu-toolchain/build/../gcc/configure --target=riscv64-unkno
nguages=c,c++ --with-pkgversion=g5115c7e447 --with-system-zlib --enable-tls --with-newlib --with-sysro
e --disable-libmudflap --disable-libssp --disable-libquadmath --disable-libgomp --disable-nls --disabl
h=rv64gc --with-isa-spec=20191213 'CFLAGS_FOR_TARGET=-Os    -mcmodel=medlow' 'CXXFLAGS_FOR_TARGET=-Os
Thread model: single
Supported LTO compression algorithms: zlib
gcc version 15.2.0 (g5115c7e447)
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain/build$ which riscv64-unknown-elf-gcc
/home/deepesh/RISCV/bin/riscv64-unknown-elf-gcc
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain/build$
```

Enter Proxy Kernel Source and Create Build Directory:

cd ..
cd ..
cd riscv-pk
mkdir build
cd build

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain/build$ cd ..
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-gnu-toolchain$ cd ..
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$ cd riscv-pk
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-pk$ mkdir build
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-pk$ cd build
```

Configure Proxy Kernel for RISC-V Target:

../configure --prefix=$RISCV --host=riscv64-unknown-elf

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-pk/build$ ../configure --prefix=$RISCV --host=riscv64-unknown-elf
checking build system type... x86_64-pc-linux-gnu
checking host system type... riscv64-unknown-elf
checking for riscv64-unknown-elf-gcc... riscv64-unknown-elf-gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... yes
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether riscv64-unknown-elf-gcc accepts -g... yes
checking for riscv64-unknown-elf-gcc option to enable C11 features... none needed
checking for riscv64-unknown-elf-g++... riscv64-unknown-elf-g++
checking whether the compiler supports GNU C++... yes
checking whether riscv64-unknown-elf-g++ accepts -g... yes
checking for riscv64-unknown-elf-g++ option to enable C++11 features... none needed
```

Build the Project Using Make:

make

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-pk/build$ make
riscv64-unknown-elf-gcc -MMD -MP -Wall -Werror -D__NO_INLINE__ -mcmodel=medany -O2 -std=gnu99 -Wno-unused -Wno-attribut
bbl_logo_file\" -DMEM_START=0x80000000 -fno-stack-protector -U_FORTIFY_SOURCE -DBBL_PAYLOAD=\"bbl_payload\" -I. -I../pk
l -c ../pk/file.c
riscv64-unknown-elf-gcc -MMD -MP -Wall -Werror -D__NO_INLINE__ -mcmodel=medany -O2 -std=gnu99 -Wno-unused -Wno-attribut
bbl_logo_file\" -DMEM_START=0x80000000 -fno-stack-protector -U_FORTIFY_SOURCE -DBBL_PAYLOAD=\"bbl_payload\" -I. -I../pk
l -c ../pk/syscall.c
riscv64-unknown-elf-gcc -MMD -MP -Wall -Werror -D__NO_INLINE__ -mcmodel=medany -O2 -std=gnu99 -Wno-unused -Wno-attribut
bbl_logo_file\" -DMEM_START=0x80000000 -fno-stack-protector -U_FORTIFY_SOURCE -DBBL_PAYLOAD=\"bbl_payload\" -I. -I../pk
l -c ../pk/handlers.c
riscv64-unknown-elf-gcc -MMD -MP -Wall -Werror -D__NO_INLINE__ -mcmodel=medany -O2 -std=gnu99 -Wno-unused -Wno-attribut
bbl_logo_file\" -DMEM_START=0x80000000 -fno-stack-protector -U_FORTIFY_SOURCE -DBBL_PAYLOAD=\"bbl_payload\" -I. -I../pk
l -c ../pk/frontend.c
riscv64-unknown-elf-gcc -MMD -MP -Wall -Werror -D__NO_INLINE__ -mcmodel=medany -O2 -std=gnu99 -Wno-unused -Wno-attribut
bbl_logo_file\" -DMEM_START=0x80000000 -fno-stack-protector -U_FORTIFY_SOURCE -DBBL_PAYLOAD=\"bbl_payload\" -I. -I../pk
l -c ../pk/elf.c
```

Install the Built Project:

make install

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-pk/build$ make install
../scripts/mk-install-dirs.sh //home/deepesh/RISCV/riscv64-unknown-elf/include/riscv-pk
mkdir //home/deepesh/RISCV/riscv64-unknown-elf/include/riscv-pk
for file in ../pk/boot.h ../pk/elf.h ../pk/file.h ../pk/frontend.h ../pk/mmap.h ../pk/pk.h ../pk/syscall.h ../pk/userm
 ../softfloat/platform.h ../softfloat/primitives.h ../softfloat/primitiveTypes.h ../softfloat/softfloat.h ../softfloat
../machine/bits.h ../machine/fdt.h ../machine/emulation.h ../machine/encoding.h ../machine/fp_emulation.h ../machine/h
achine/uart16550.h ../machine/uart_litex.h ../machine/finisher.h ../machine/unprivileged_memory.h ../machine/vm.h ; \
  do \
    /usr/bin/install -c -m 644 $file //home/deepesh/RISCV/riscv64-unknown-elf/include/riscv-pk; \
  done
../scripts/mk-install-dirs.sh //home/deepesh/RISCV/riscv64-unknown-elf/lib/riscv-pk
mkdir //home/deepesh/RISCV/riscv64-unknown-elf/lib/riscv-pk
for file in libpk.a libbbl.a libsoftfloat.a libdummy_payload.a libmachine.a libutil.a; \
  do \
    /usr/bin/install -c -m 644 $file //home/deepesh/RISCV/riscv64-unknown-elf/lib/riscv-pk; \
  done
```

Enter Spike Source and Create Build Directory:

cd ..
cd ..
cd riscv-isa-sim

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-pk/build$ cd ..
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-pk$ cd ..
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$ cd riscv-isa-sim
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim$
```

mkdir build
cd build

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim$ mkdir build
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim$ cd build
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim/build$
```

Configure Spike Build for Installation Directory:

../configure --prefix=$RISCV

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim/build$ ../configure --prefix=$RISCV
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes
checking for gcc option to enable C11 features... none needed
checking for g++... g++
checking whether the compiler supports GNU C++... yes
checking whether g++ accepts -g... yes
checking for g++ option to enable C++11 features... none needed
```

Build the Project Using Make:

make

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim/build$ make
g++ -MMD -MP  -DPREFIX=\"/home/deepesh/RISCV\" -Wall -Wno-nonportable-include-path -g -
iquote ../customext -iquote ../fdt -iquote ../softfloat -iquote ../spike_main -iquote .
g++ -MMD -MP  -DPREFIX=\"/home/deepesh/RISCV\" -Wall -Wno-nonportable-include-path -g -
iquote ../customext -iquote ../fdt -iquote ../softfloat -iquote ../spike_main -iquote .
g++ -MMD -MP  -DPREFIX=\"/home/deepesh/RISCV\" -Wall -Wno-nonportable-include-path -g -
iquote ../customext -iquote ../fdt -iquote ../softfloat -iquote ../spike_main -iquote .
g++ -MMD -MP  -DPREFIX=\"/home/deepesh/RISCV\" -Wall -Wno-nonportable-include-path -g -
iquote ../customext -iquote ../fdt -iquote ../softfloat -iquote ../spike_main -iquote .
g++ -MMD -MP  -DPREFIX=\"/home/deepesh/RISCV\" -Wall -Wno-nonportable-include-path -g -
iquote ../customext -iquote ../fdt -iquote ../softfloat -iquote ../spike_main -iquote .
g++ -MMD -MP  -DPREFIX=\"/home/deepesh/RISCV\" -Wall -Wno-nonportable-include-path -g -
iquote ../customext -iquote ../fdt -iquote ../softfloat -iquote ../spike_main -iquote .
g++ -MMD -MP  -DPREFIX=\"/home/deepesh/RISCV\" -Wall -Wno-nonportable-include-path -g -
iquote ../customext -iquote ../fdt -iquote ../softfloat -iquote ../spike_main -iquote .
g++ -MMD -MP  -DPREFIX=\"/home/deepesh/RISCV\" -Wall -Wno-nonportable-include-path -g -
iquote ../customext -iquote ../fdt -iquote ../softfloat -iquote ../spike_main -iquote .
```

Install the Built Project:

make install

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim/build$ make install
../scripts/mk-install-dirs.sh /home/deepesh/RISCV/include
for file in fesvr/byteorder.h fesvr/elf.h fesvr/elfloader.h fesvr/htif.h fesvr/dtm.h fe
ion_parser.h fesvr/term.h fesvr/device.h fesvr/rfb.h fesvr/tsi.h riscv/abstract_device.
ommon.h riscv/csrs.h riscv/debug_defines.h riscv/debug_module.h riscv/debug_rom_defines
e.h riscv/extension.h riscv/isa_parser.h riscv/jtag_dtm.h riscv/log_file.h riscv/memtra
m.h riscv/simif.h riscv/trap.h riscv/triggers.h riscv/vector_unit.h fdt/libfdt.h fdt/fd
  do \
        ../scripts/mk-install-dirs.sh /home/deepesh/RISCV/include/`dirname $file`; \
    /usr/bin/install -c -m 644 ../$file /home/deepesh/RISCV/include/`dirname $file`; \
  done
mkdir /home/deepesh/RISCV/include/fesvr
mkdir /home/deepesh/RISCV/include/riscv
mkdir /home/deepesh/RISCV/include/fdt
mkdir /home/deepesh/RISCV/include/softfloat
../scripts/mk-install-dirs.sh /home/deepesh/RISCV/include
for dir in fesvr       ; \
  do \
        ../scripts/mk-install-dirs.sh /home/deepesh/RISCV/include/$dir; \
```

Verify Spike Installation:

spike --help

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim/build$ spike --help
Spike RISC-V ISA Simulator 1.1.1-dev

usage: spike [host options] <target program> [target options]
Host Options:
  -p<n>                 Simulate <n> processors [default 1]
  -m<n>                 Provide <n> MiB of target memory [default 2048]
  -m<a:m,b:n,...>       Provide memory regions of size m and n bytes
                          at base addresses a and b (with 4 KiB alignment)
  -d                    Interactive debug mode
  -g                    Track histogram of PCs
  -l                    Generate a log of execution
  -h, --help            Print this help message
  --halted              Start halted, allowing a debugger to connect
  --log=<name>          File name for option -l
  --debug-cmd=<name>    Read commands from file (use with -d)
  --isa=<name>          RISC-V ISA string [default rv64imafdc_zicntr_zihpm]
  --pmpregions=<n>      Number of PMP regions [default 16]
  --pmpgranularity=<n>  PMP Granularity in bytes [default 4]
```

Create Directory for COA Programs:

cd ..
cd ..
cd ..
mkdir COA-programs
cd COA-programs

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim/build$ cd ..
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV/riscv-isa-sim$ cd ..
deepesh@deepesh-HP-Laptop-15-da0xxx:~/RISCV$ cd ..
deepesh@deepesh-HP-Laptop-15-da0xxx:~$ mkdir COA-programs
deepesh@deepesh-HP-Laptop-15-da0xxx:~$ cd COA-programs/
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$
```

Create a New RISC-V C Program:

nano pro1_hello.c

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$ nano pro1_hello.c
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$
```

```
  GNU nano 7.2
#include<stdio.h>
int main()
{
        printf("Hello RISCV !\n");
        return 0;
}
```

Press Ctrl + O → press Enter to save

Press Ctrl + X → to exit the editor

Compile RISC-V Program and Verify Output:

riscv64-unknown-elf-gcc pro1_hello.c

ls -l

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$ riscv64-unknown-elf-gcc pro1_hello.c
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$ ls -l
total 28
-rwxrwxr-x 1 deepesh deepesh 23544 Jan  6 09:50 a.out
-rw-rw-r-- 1 deepesh deepesh    75 Jan  6 09:50 pro1_hello.c
```

Verify the RISC-V Binary:

file a.out

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$ file a.out
a.out: ELF 64-bit LSB executable, UCB RISC-V, RVC, double-float ABI, version 1 (SYSV), statically linked, not stripped
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$
```

Run the RISC-V Program on Spike Using Proxy Kernel:

spike pk a.out

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$ spike pk a.out
Hello RISCV !
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$
```

Generate RISC-V Assembly from C Source:

riscv64-unknown-elf-gcc -S pro1_hello.c -o pro1_hello.s

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$ riscv64-unknown-elf-gcc -S pro1_hello.c -o pro1_hello.s
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$ ls -l
total 32
-rwxrwxr-x 1 deepesh deepesh 23544 Jan  6 09:50 a.out
-rw-rw-r-- 1 deepesh deepesh    75 Jan  6 09:50 pro1_hello.c
-rw-rw-r-- 1 deepesh deepesh   609 Jan  6 11:08 pro1_hello.s
```

View the RISC-V Assembly Code:

less pro1_hello.s

```
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$ less pro1_hello.s
deepesh@deepesh-HP-Laptop-15-da0xxx:~/COA-programs$
```

```
        .file   "pro1_hello.c"
        .option nopic
        .attribute arch, "rv64i2p1_m2p0_a2p1_f2p2_d2p2_c2p0_zicsr2p0_zifencei2p0_zmmul1p0_zaamo1p0_zalrsc1p0_zca1p0_zcd1p0"
        .attribute unaligned_access, 0
        .attribute stack_align, 16
        .text
        .section        .rodata
        .align  3
.LC0:
        .string "Hello RISCV !"
        .text
        .align  1
        .globl  main
        .type   main, @function
main:
        addi    sp,sp,-16
        sd      ra,8(sp)
        sd      s0,0(sp)
        addi    s0,sp,16
        lui     a5,%hi(.LC0)
        addi    a0,a5,%lo(.LC0)
        call    puts
        li      a5,0
        mv      a0,a5
        ld      ra,8(sp)
        ld      s0,0(sp)
        addi    sp,sp,16
        jr      ra
        .size   main, .-main
        .ident  "GCC: (g5115c7e447) 15.2.0"
        .section        .note.GNU-stack,"",@progbits
pro1_hello.s (END)
```

Note-

- Write simple c programs for addition, subtraction, multiplication and division.
- Execute it like above hello program.
- Try to understand the corresponding generated assembly code.